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# Training/Testing 80/20
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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics import accuracy_score, precision_score,
recall_score, f1_score, confusion_matrix

# =====
# 1. Load Dataset
# =====
df = pd.read_csv("/content/spam_or_not_spam.csv")
df.dropna(subset=['email', 'label'], inplace=True)

X = df['email']
y = df['label'].values.reshape(-1,1)

# =====
# 2. Train-Test Split
# =====
X_train_text, X_test_text, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# =====
# 3. TF-IDF
# =====
vectorizer = TfidfVectorizer(stop_words='english', max_features=2000)
X_train = vectorizer.fit_transform(X_train_text).toarray()
X_test = vectorizer.transform(X_test_text).toarray()

# =====
# 4. Network Architecture
# =====
input_size = X_train.shape[1]
h1 = 64
h2 = 32
output_size = 1
epochs = 200

learning_rates = [0.01, 0.005, 0.001]

# =====
# Activation Functions
# =====
```

```

def relu(x):
    return np.maximum(0, x)

def relu_derivative(x):
    return (x > 0).astype(float)

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

loss_results = {}
accuracy_results = {}

# =====
# 5. Train for Each LR
# =====
for lr in learning_rates:

    print("\n=====")
    print(f"Training with Learning Rate = {lr}")
    print("=====")
    np.random.seed(42)

    W1 = np.random.randn(input_size, h1) * 0.01
    b1 = np.zeros((1, h1))

    W2 = np.random.randn(h1, h2) * 0.01
    b2 = np.zeros((1, h2))

    W3 = np.random.randn(h2, output_size) * 0.01
    b3 = np.zeros((1, output_size))

    loss_list = []
    accuracy_list = []

    for epoch in range(epochs):

        # Forward
        Z1 = X_train.dot(W1) + b1
        A1 = relu(Z1)

        Z2 = A1.dot(W2) + b2
        A2 = relu(Z2)

        Z3 = A2.dot(W3) + b3
        A3 = sigmoid(Z3)

        # BCE Loss
        loss = -np.mean(y_train*np.log(A3+1e-9) + (1-
y_train)*np.log(1-A3+1e-9))
        loss_list.append(loss)

```

```

# Training Accuracy
preds_train = (A3 >= 0.5).astype(int)
acc = accuracy_score(y_train, preds_train)
accuracy_list.append(acc)

# Backprop
dZ3 = A3 - y_train
dW3 = A2.T.dot(dZ3) / len(X_train)
db3 = np.sum(dZ3, axis=0, keepdims=True) / len(X_train)

dA2 = dZ3.dot(W3.T)
dZ2 = dA2 * relu_derivative(Z2)
dW2 = A1.T.dot(dZ2) / len(X_train)
db2 = np.sum(dZ2, axis=0, keepdims=True) / len(X_train)

dA1 = dZ2.dot(W2.T)
dZ1 = dA1 * relu_derivative(Z1)
dW1 = X_train.T.dot(dZ1) / len(X_train)
db1 = np.sum(dZ1, axis=0, keepdims=True) / len(X_train)

# Update
W3 -= lr * dW3
b3 -= lr * db3
W2 -= lr * dW2
b2 -= lr * db2
W1 -= lr * dW1
b1 -= lr * db1

loss_results[lr] = loss_list
accuracy_results[lr] = accuracy_list

# =====
# FINAL TEST EVALUATION
# =====
Z1 = X_test.dot(W1) + b1
A1 = relu(Z1)

Z2 = A1.dot(W2) + b2
A2 = relu(Z2)

Z3 = A2.dot(W3) + b3
A3 = sigmoid(Z3)

test_preds = (A3 >= 0.5).astype(int)

print("\nFinal Test Metrics:")
print("Accuracy :", accuracy_score(y_test, test_preds))
print("Precision:", precision_score(y_test, test_preds))
print("Recall    :", recall_score(y_test, test_preds))
print("F1 Score  :", f1_score(y_test, test_preds))

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    print("Confusion Matrix:\n", confusion_matrix(y_test, test_preds))

# =====
# 6. Plot Loss Comparison
# =====
plt.figure()
for lr in learning_rates:
    plt.plot(range(1, epochs+1), loss_results[lr])

plt.title("Loss vs Epochs (MLP)")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.legend([f"LR = {lr}" for lr in learning_rates])
plt.show()

# =====
# 7. Plot Accuracy Comparison
# =====
plt.figure()
for lr in learning_rates:
    plt.plot(range(1, epochs+1), accuracy_results[lr])

plt.title("Accuracy vs Epochs (MLP)")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend([f"LR = {lr}" for lr in learning_rates])
plt.show()

=====
Training with Learning Rate = 0.01
=====

Final Test Metrics:
Accuracy : 0.8333333333333334
Precision: 0.0
Recall   : 0.0
F1 Score : 0.0
Confusion Matrix:
[[500  0]
 [100  0]]

=====
Training with Learning Rate = 0.005
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/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 due to no predicted samples. Use

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`zero_division` parameter to control this behavior.  
_warn_prf(average, modifier, f"{metric.capitalize()} is",  
len(result))
```

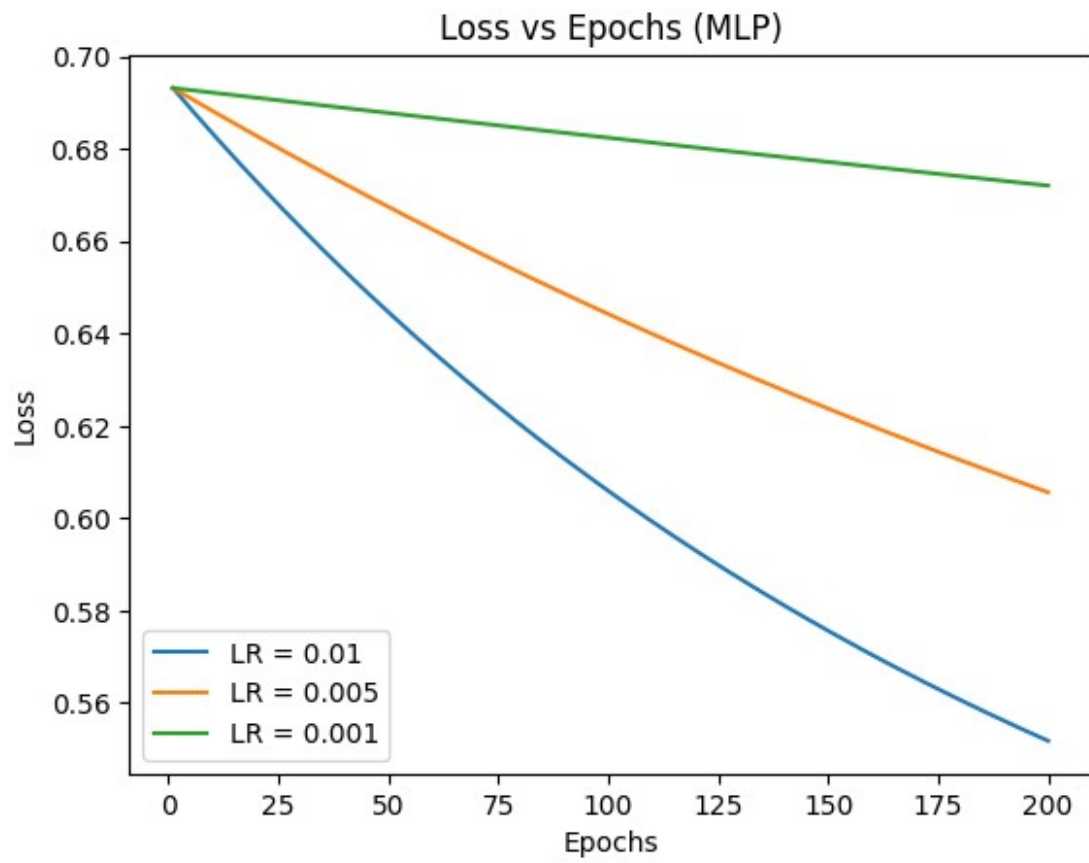
```
Final Test Metrics:  
Accuracy : 0.8333333333333334  
Precision: 0.0  
Recall    : 0.0  
F1 Score  : 0.0  
Confusion Matrix:  
[[500   0]  
 [100   0]]
```

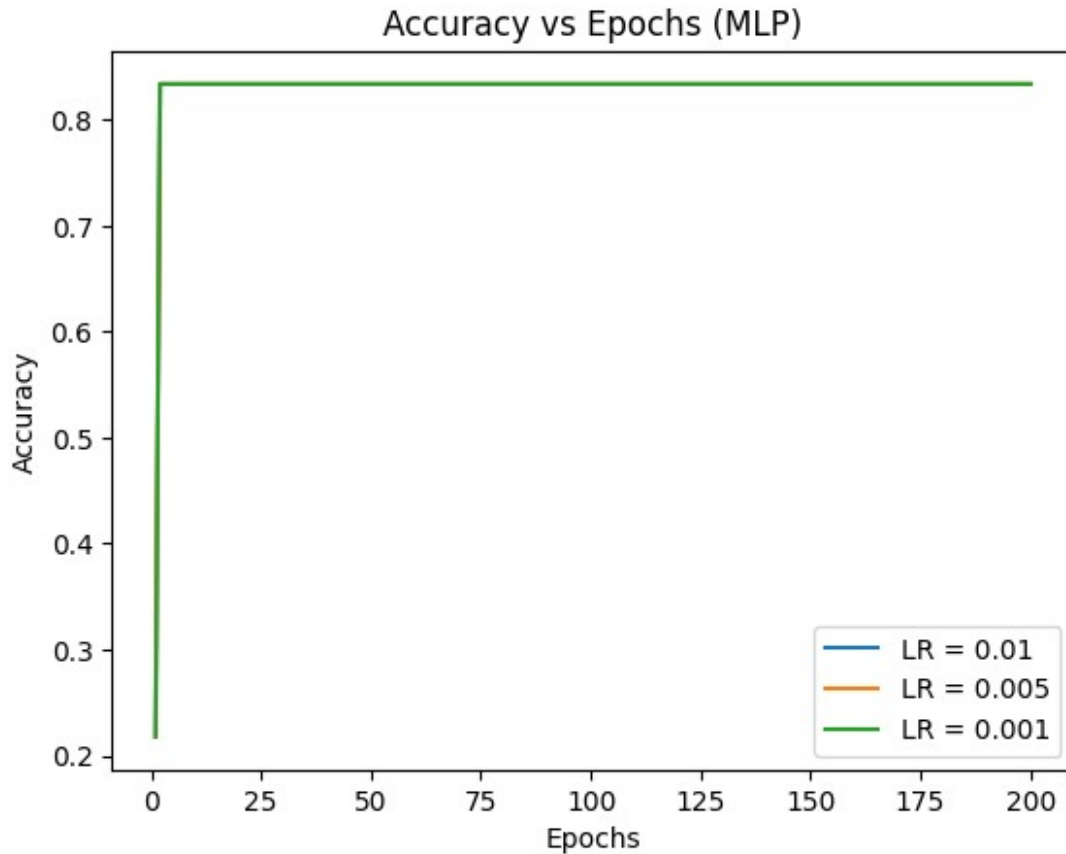
```
=====  
Training with Learning Rate = 0.001  
=====
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/  
_classification.py:1565: UndefinedMetricWarning: Precision is ill-  
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Final Test Metrics:  
Accuracy : 0.8333333333333334  
Precision: 0.0  
Recall    : 0.0  
F1 Score  : 0.0  
Confusion Matrix:  
[[500   0]  
 [100   0]]
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df.dropna(subset=['email', 'label'], inplace=True)

X = df['email']
y = df['label'].values.reshape(-1,1)

# =====
# 2. Train-Test Split
# =====
X_train_text, X_test_text, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```

# =====
# 3. TF-IDF
# =====
vectorizer = TfidfVectorizer(stop_words='english', max_features=2000)
X_train = vectorizer.fit_transform(X_train_text).toarray()
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# =====
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for lr in learning_rates:

    print("\n=====")
    print(f"Training with Learning Rate = {lr}")
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    np.random.seed(42)

    W1 = np.random.randn(input_size, h1) * 0.01
    b1 = np.zeros((1, h1))

    W2 = np.random.randn(h1, h2) * 0.01
    b2 = np.zeros((1, h2))

```



```

W3 = np.random.randn(h2, output_size) * 0.01
b3 = np.zeros((1, output_size))

loss_list = []
accuracy_list = []

for epoch in range(epochs):
    # Forward
    Z1 = X_train.dot(W1) + b1
    A1 = relu(Z1)

    Z2 = A1.dot(W2) + b2
    A2 = relu(Z2)

    Z3 = A2.dot(W3) + b3
    A3 = sigmoid(Z3)

    # BCE Loss
    loss = -np.mean(y_train*np.log(A3+1e-9) + (1-
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    dZ3 = A3 - y_train
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    db3 = np.sum(dZ3, axis=0, keepdims=True) / len(X_train)

    dA2 = dZ3.dot(W3.T)
    dZ2 = dA2 * relu_derivative(Z2)
    dW2 = A1.T.dot(dZ2) / len(X_train)
    db2 = np.sum(dZ2, axis=0, keepdims=True) / len(X_train)

    dA1 = dZ2.dot(W2.T)
    dZ1 = dA1 * relu_derivative(Z1)
    dW1 = X_train.T.dot(dZ1) / len(X_train)
    db1 = np.sum(dZ1, axis=0, keepdims=True) / len(X_train)

    # Update
    W3 -= lr * dW3
    b3 -= lr * db3
    W2 -= lr * dW2
    b2 -= lr * db2
    W1 -= lr * dW1
    b1 -= lr * db1

```

```

loss_results[lr] = loss_list
accuracy_results[lr] = accuracy_list

# =====
# FINAL TEST EVALUATION
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Z1 = X_test.dot(W1) + b1
A1 = relu(Z1)

Z2 = A1.dot(W2) + b2
A2 = relu(Z2)

Z3 = A2.dot(W3) + b3
A3 = sigmoid(Z3)

test_preds = (A3 >= 0.5).astype(int)

print("\nFinal Test Metrics:")
print("Accuracy :", accuracy_score(y_test, test_preds))
print("Precision:", precision_score(y_test, test_preds))
print("Recall    :", recall_score(y_test, test_preds))
print("F1 Score  :", f1_score(y_test, test_preds))
print("Confusion Matrix:\n", confusion_matrix(y_test, test_preds))

# =====
# 6. Plot Loss Comparison
# =====
plt.figure()
for lr in learning_rates:
    plt.plot(range(1, epochs+1), loss_results[lr])

plt.title("Loss vs Epochs (MLP)")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.legend([f"LR = {lr}" for lr in learning_rates])
plt.show()

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    plt.plot(range(1, epochs+1), accuracy_results[lr])

plt.title("Accuracy vs Epochs (MLP)")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend([f"LR = {lr}" for lr in learning_rates])
plt.show()

```

```
=====
Training with Learning Rate = 0.01
=====
```

```
Final Test Metrics:
Accuracy : 0.8233333333333334
Precision: 0.0
Recall   : 0.0
F1 Score : 0.0
Confusion Matrix:
[[741  0]
 [159  0]]
```

```
=====
Training with Learning Rate = 0.005
=====
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 due to no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
Final Test Metrics:
Accuracy : 0.8233333333333334
Precision: 0.0
Recall   : 0.0
F1 Score : 0.0
Confusion Matrix:
[[741  0]
 [159  0]]
```

```
=====
Training with Learning Rate = 0.001
=====
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 due to no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
Final Test Metrics:
Accuracy : 0.8233333333333334
Precision: 0.0
Recall   : 0.0
```

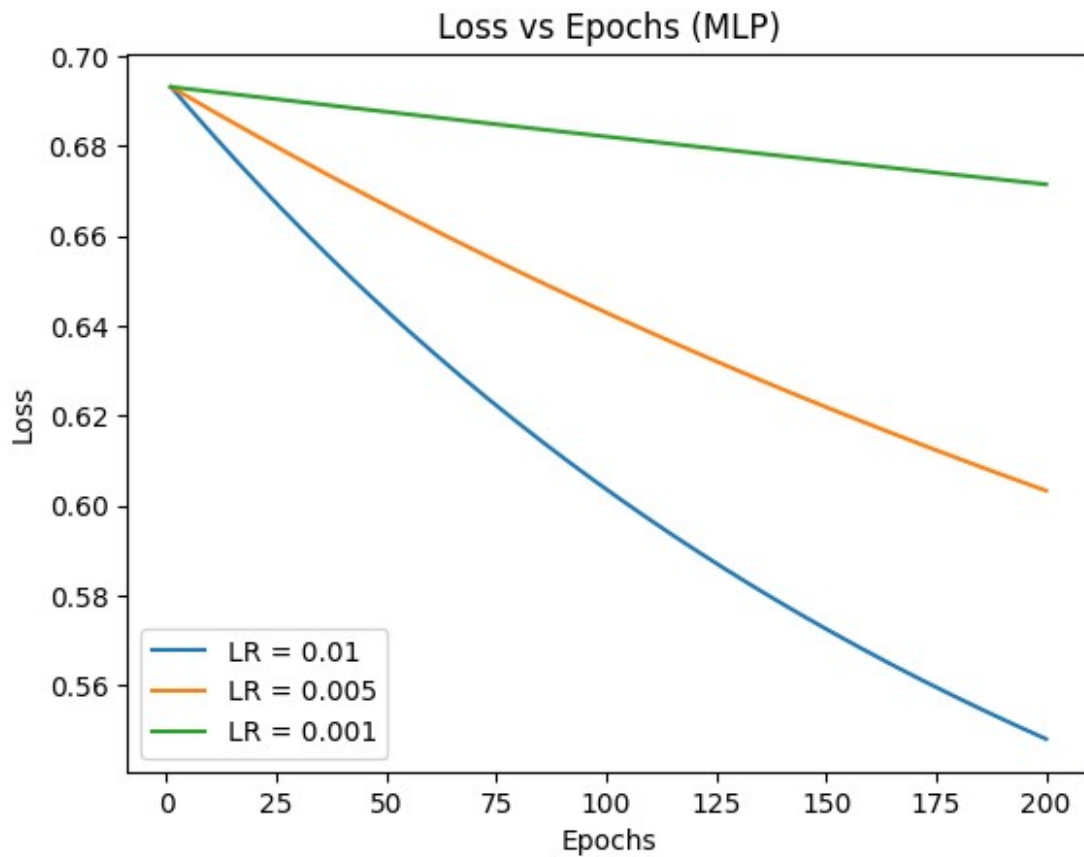
F1 Score : 0.0

Confusion Matrix:

```
[[741  0]
```

```
[159  0]]
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 due to no predicted samples. Use `zero_division` parameter to control this behavior.  
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```



Accuracy vs Epochs (MLP)

